

NUS “SHOW ME YOUR AGENTS” HACKATHON

Why an Agent Council? — Sumin’s Research & Product Thesis

Smart Farm Production Planning • Pre-Meeting Brief • Sep 9, 2026

01 — The idea in one sentence

Prediction tells us what might happen. An Agent Council helps us decide what to do about it.

Our thesis: farm production planning should not be framed as a single prediction task. It is a multi-constraint decision problem where demand, weather, production capacity, supply chain, price and profitability can disagree.

02 — Why did we think of this?

At first glance, the hackathon problem sounds like a classic ML problem: predict demand or crop production and output a number. But that is not how an actual farm manager experiences the problem.

- A demand forecast can say: “Customers may need 1,000 kg of tomatoes.”
- A weather model can say: “Heavy rainfall creates yield risk.”
- A market model can say: “Tomato prices may fall.”
- A production model can say: “The farm can physically produce 1,200 kg.”
- A finance model can say: “Another crop has a better expected margin.”

None of these answers is wrong. The difficult part is deciding what to do when they conflict. That is why our system needs multiple specialist agents and one planning agent that resolves trade-offs.

03 — The ‘Hong Sang-soo’ test

A strange-looking idea is not necessarily a bad idea.

Think of a Hong Sang-soo film: the surface can look deliberately simple, repetitive or even strange to someone seeing it for the first time, yet that unusual structure can be precisely what makes the work distinctive and internationally recognized. The lesson for our hackathon is not that we should make something weird for the sake of being weird. It is that judges should understand the reason behind an unconventional design.

Our unconventional choice is not “let’s use many agents because agents are trendy.” Our choice is: “the real-world decision itself contains multiple expert viewpoints, so we model those viewpoints explicitly.”

The architecture is the argument.

04 — The Singapore reality makes this problem sharper

Singapore is a particularly good context for this idea. The government has explicitly identified the need to match local farm supply with market demand. SAFEF acts as a demand-aggregation engine that connects fragmented farm production with buyers, while longer-term offtake arrangements give farms greater demand certainty.

In 2026, MSE described demand aggregation as central to helping farms scale: buyers want consistent supply, while farms need confidence that what they produce can be sold. This is almost exactly the coordination problem our Agent Council is designed to represent.

Sources: MSE 2026; MSE 2024. [citeturn0search1turn0search0](#)

05 — Reframing the problem

Typical AI framing

Our agentic framing

Predict crop demand	Coordinate competing signals into a production decision
One model	A council of specialist perspectives
Output = number	Output = plan + rationale + risks
Static prediction	Interactive scenario planning
AI replaces the farmer	AI supports the planning organization

06 — The Agent Council

Agent	Core question	Inputs	Output
Demand Agent	What will buyers need?	Demand history, seasonality, market signals	Demand forecast
Weather Agent	What environmental risk is coming?	Rainfall, temperature, humidity, forecasts	Risk / yield adjustment
Market Agent	What can we sell, and at what price?	Price and market signals	Price / attractiveness
Production Agent	What can the farm actually grow?	Land, crop cycle, capacity, yield	Feasible quantity
Supply Chain Agent	Can we support the plan?	Inputs, inventory, logistics	Constraints / alerts
Profit Agent	Which option makes economic sense?	Cost, yield, price, demand	Margin / ranking
Planner Agent	So what should we do?	All agent outputs	Final plan + explanation

07 — Why the Planner Agent matters

The Planner Agent should not simply average the other agents. It should explain trade-offs and make a constrained recommendation.

Example:

Demand Agent: tomato demand –15%

Weather Agent: rainfall risk +20%

Market Agent: tomato price expected –8%

Production Agent: capacity can be reallocated

Profit Agent: leafy greens have higher expected margin

Planner Agent → “Reduce tomato production by 20%, reallocate part of capacity to leafy greens, and retain a safety buffer because rainfall uncertainty remains high.”

08 — Make the demo interactive, not just pretty

The gamification requirement becomes much more meaningful if judges can change one variable and watch the agent organization react.

- Slider: Tomato demand ↓ 15%
- Weather toggle: Heavy rainfall ON
- Price shock: Tomato price –10%
- Supply constraint: Fertilizer availability –15%

The screen can show six specialist cards debating the scenario, then a Planner Agent producing the revised farm plan. This makes the agent collaboration visible rather than hiding everything behind one chatbot answer.

09 — Data strategy: don’t force incompatible resolutions

A major technical risk identified by the team is the mismatch between annual/seasonal production data and hourly weather observations. We should not join them naively. Instead, engineer weather features at the planning horizon: daily/weekly averages, cumulative rainfall, anomalies, extreme-weather flags, and rolling windows such as rainfall_7d and rainfall_30d.

Singapore's open-data ecosystem provides a practical starting point, while the exact crop/production dataset should be chosen only after the official problem statement is confirmed.

10 — What makes this hackathon-worthy?

- Clear thesis: we model an organization, not just a prediction.
- Agent roles have business meaning, not arbitrary personas.
- Agents can disagree, creating a visible reason for orchestration.
- The Planner Agent converts predictions into an actionable decision.
- Interactive scenarios demonstrate adaptation in real time.
- The Singapore supply-demand context makes the problem locally relevant.

11 — What we should NOT build

- A generic farming chatbot.
- Seven agents that all call the same LLM and return vague text.
- A beautiful dashboard with no working decision loop.
- A huge ML pipeline that cannot be explained to judges.
- A literal farming game that distracts from the production-planning challenge.

12 — The 90-second story for judges

“At first, we thought farm production planning was a forecasting problem. Then we realized that knowing what will happen is not enough. A farmer still has to decide what to do when demand, weather, capacity, price and profitability disagree. So instead of building one AI predictor, we built an AI organization. Our Demand,

Weather, Market, Production, Supply Chain and Profit Agents each represent a different operational perspective. The Planner Agent acts like the decision-maker, resolving their trade-offs into one production plan. The interface lets users change a scenario and watch the organization adapt. We are not building an AI that predicts what a farm should grow. We are building an AI organization that debates what a farm should grow.”

13 — Wednesday meeting: decisions we need

- . ☐ Confirm the official SBF problem statement and judging criteria.
- . ☐ Choose 3–5 crops and one planning horizon.
- . ☐ Agree which agents are essential for the MVP.
- . ☐ Decide which outputs are ML predictions vs. rules/LLM reasoning.
- . ☐ Select the minimum viable datasets.
- . ☐ Choose one dramatic interactive scenario for the final demo.
- . ☐ Assign agent orchestration, AWS/Cloudflare, frontend and pitch ownership.

14 — Sumin’s concrete contribution

Before the meeting, Sumin should own the business thesis and agent logic: why the problem requires multiple perspectives, what each agent represents, what information each agent consumes, and how the Planner Agent turns disagreement into an actionable plan.

Key line to bring into the meeting: “Let’s make the agents disagree on purpose. If they never disagree, we don’t need an agent council.”

15 — Sources & evidence

. MSE — Committee of Supply 2026: demand aggregation, local farm economics and supply certainty —

<https://www.mse.gov.sg/latest-news/speech-by-sms-zaqy-mohamad---committee-of-supply-2026/>

. MSE — Launch of Aggregated Local Produce at FairPrice, 2024 —

<https://www.mse.gov.sg/latest-news/launch-of-aggregated-local-produce/>

. MSE — Written Reply on SAFEF collaboration, 2025 —

<https://www.mse.gov.sg/latest-news/written-reply-to-parliamentary-question-on-collaboration-with-safef/>

. MSE — Market Access for Local Farms, 2026 —

<https://www.mse.gov.sg/latest-news/written-reply-to-parliamentary-question-on-market-access-for-local-farms/>

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